

YEAR 9 SCIENCE CHEMISTRY

Aim: To determine the pH of some common substances.

Materials: Porcelain spotter tiles
Universal indicator + colour charts
Test substances - white vinegar, lemon juice, orange juice, detergent, disinfectant, soda water, soap solution, bleach, milk, milk of magnesia, toothpaste, sodium chloride solution, swimming pool water, health salts or substitute, drain cleaner, 0.1 M HCl, 0.1 M NaOH.

Method:

1. Place a small amount of the test substance into a depression on the spotter tile.
2. Add 1-2 drops of universal indicator and use the colour chart to determine the pH.
3. Record the values in a table and determine if the substance is an acid (strong or weak), base (strong or weak) or neutral

Results:

[illegible]

Discussion: Answer the following questions.

1. What type of substances are acidic? Is there any pattern?
2. What type of substances are bases? Is there any pattern?
3. Were any substances difficult to test?
4. Were there any problems doing the tests? Do you have any suggestions for improvements?

Conclusion: How successful were you in determining the pH of substances?